

Program : Diploma in Wood and Paper Technology	
Course Code : 5123B	Course Title: Quantity Surveying
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To provide knowledge on Quantity surveying and measuring principles
- To impart knowledge on Price Analysis
- To familiarize valuation of land and building
- To introduce computer applications in Quantity surveying

Course Prerequisites:

Topic	Course code	Course Title	Semester
Knowledge of basic Mathematics		Mathematics I	1
Knowledge in Statistics, Mean, Median, Mode		Mathematics II	2

Course Outcome:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Recall Quantity Surveying and discuss the principles of measurements and description	15	Understanding
CO2	Comprehend analysis of price and understand tendering and contracting	14	Applying
CO3	Summarize the land evaluation and surveyor laws	15	Applying
CO4	Describe Project management methods applications	14	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2					3		
CO3	3						
CO4						3	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Recall Quantity Surveying and discuss principles of measurements and description.		
M1.01	Explain basics of quantity surveying	5	Remembering
M1.02	Explain building estimates	5	Remembering
M1.03	Develop an idea on measurement of work	5	Understanding
Contents: Historical development of quantity surveying: Functions performed by the quantity surveyor in relation to construction works - Evolution of standard methods of measurement for construction Works. The use of document in practice. Basic concepts of quantity surveying: Measurement and computation of lengths, girths, areas, and volumes both for regular and irregular shapes from drawings. Theoretical processes of building contract from inception to completion and the interrelationship of the professional team Engineering drawings for buildings and various structures, Principles of standard methods of measurement of work. Bills of quantities, Drawings. Purpose of Bills of Quantities. Processes of preparing Bills of Quantities, including taking off, working up, abstracting and billing. Types of bill formats and their uses. Setting out of descriptive and quantitative information in taking off. Dimension, abstracting and Billing Sheets			
CO2	Comprehend analysis of Price and understand tendering and contracting		
M2.01	Explain price analysis of labours and construction	5	Understanding
M2.02	Explain estimation and contracting and tendering	4	Applying
M2.03	Develop a solution for contract conditions and administrations	5	Understanding
	Series Test – I	1	

Contents:

Analysis of prices of Construction materials, Labor, Machinery, Overheads & incidentals, Other incidentals in the analysis of rates for various trade items of civil engineering construction, Standard schedule of rates,

Machinery and their uses in various construction activities, Output constants, Life, depreciation, cost of owning and operating charges for machinery, Economics of owning or hiring machinery.

Estimating, tendering and contract management .Tender structure for different types of contracts, Tendering procedure, Specifications, Preparation of pre-qualification documents, Preparation of Tender

Documents and invitation to bid, Submission of bids, Tender documents for LCB, NCB and ICB, Procurement of tender documents, Pre-bid conference and site investigation, Bank guarantees and Performance Bonds, Sub contracting &labour contracting system, Letter of Acceptance and Contract signing.

CO3	Summarize the land evaluation and surveyor laws		
M3.01	Explain law of surveyors	5	Remembering
M3.02	Explain the assessment of property	5	Applying
M3.03	Explain valuation of land and building	5	Applying

Contents:

Valuation - Explanation of terms, types of values, sinking fund, years purchase, Depreciation - Straight line method, constant percentage method, S.F method .Obsolescence.

Valuation of real properties-rental method, profit based method, depreciation method. Valuation of landed properties -belting method, development method, hypothecated building scheme method. Rent calculation. Lease and Lease hold property Law of Surveyors- Valuation of land, buildings and machinery.

The principal types of landed property, the incidents of their tenure, Income derived from them and the outgoing to which they are subjected; Laws governing valuation of land & building; Methods of valuation of Construction machinery.

CO4	Describe Project management methods and applications		
M4.01	Explain project management	7	Remembering
M4.02	Develop solutions for Project work breakdown	7	Applying
	Series Test – II	1	

Contents:

Project Management - functions, processes, organization. Project manager's role, responsibility, competency criteria; main causes of project failure. Project development process. Project management organization, Project planning and control approach.

Types of scheduling : Barchart, Network, Line of Balance schedule, Time limited and Resources limited schedule, Schedule hierarchy, Resources forecasting, Manpower planning, Material planning, Material Procurement schedule, Planning and selection of equipment. Cost planning and Scheduling Computer applications in quantity surveying Quantity estimation, Tendering, Billing and Project Management concept; Quick and accurate quantity calculation;

Text /Reference:

T/R	Book Title/Author
T1	Martin Brook, Estimating and tendering for construction work, Butterworth-4th edition
R2	Trever Holroyd, Principles of Estimating, Thomas Tesford, 2nd edition
R3	Aliyanashworth, Practice and Procedure for Quantity Surveyors, Wileyindia
R4	D DKohli, R CKohli. A Text book of Estimating and costing

Online resources

Sl. No	Website Link
1	https://youtu.be/hfi59uBYacg
2	https://www.quora.com/What-is-the-importance-and-purpose-of-quantity-surveying-and-costing-in-construction-work-1
3	https://engineeringmanagementinstitute.org/quantity-surveyors-why-important/
4	https://www.slideshare.net/Wallsasia/quantity-surveying-and-its-importance